

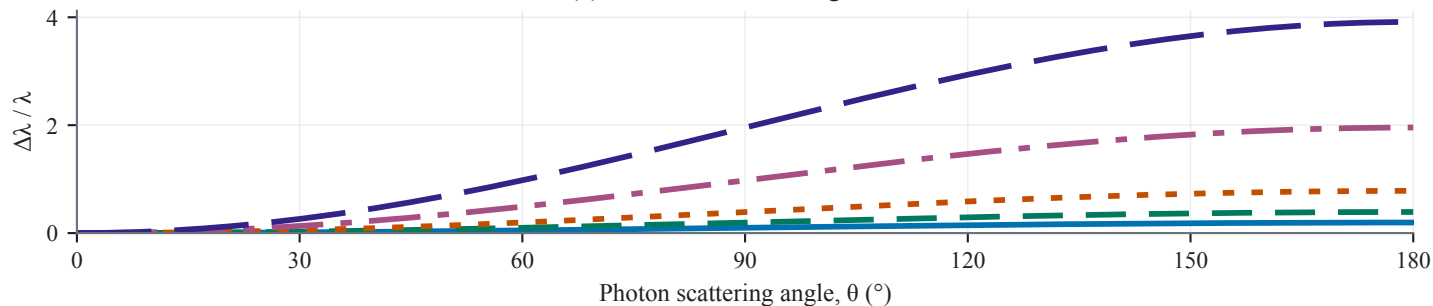
Compton scattering: the three required angular dependences

Exact relativistic free-electron model

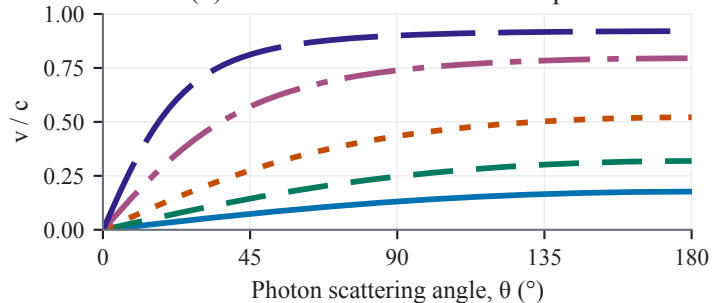
Exact relativistic free-electron kinematics · five official incident energies · 721 points from 0° to 180°

50 keV · solid 100 keV · dashed 200 keV · dotted 500 keV · dash-dot 1000 keV · long dash

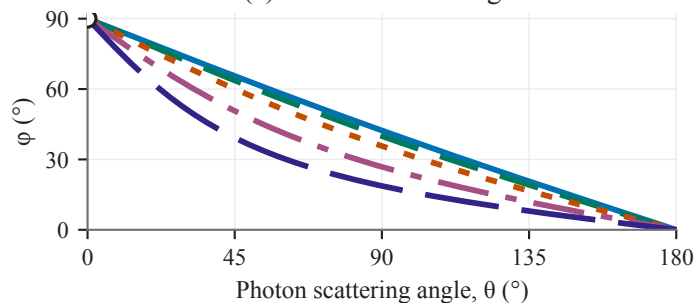
(a) Fractional wavelength shift



(b) Relativistic electron recoil speed



(c) Electron recoil angle



$\theta = 0^\circ$: zero electron momentum, so ϕ is undefined; the open point is its 90° limit. · $\theta = 180^\circ$: $\phi = 0^\circ$.